

BLUESTOCK FINTECH PVT. LTD.

Mutual Fund Analytics Platform

Capstone Project — Final Report

Project Domain:

Financial Data Engineering & Analytics

Platform:

Bluestock Fintech Pvt. Ltd.

Version:

v1.0 — Final Submission

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1. Executive Summary

The Bluestock Mutual Fund Analytics Platform is a full-stack data engineering and analytics capstone project designed to provide institutional-grade insights into India's rapidly growing mutual fund industry. Leveraging AMFI-structured datasets spanning January 2022 to May 2026, the project implements an end-to-end ETL pipeline, statistical risk-return analytics, investor behaviour profiling, and a Power BI dashboard for interactive decision support.

India's mutual fund industry has witnessed exponential growth, with SIP inflows crossing Rs 31,002 crore in December 2025 — a 17% year-on-year increase — and total folios exceeding 9.35 crore active SIP accounts. This project transforms raw AMFI data across 10 structured datasets into actionable intelligence for fund managers, retail investors, and analysts.

Key Achievements

- Processed 46,000+ NAV records and 32,778 investor transactions through a robust ETL pipeline
- Computed risk metrics including Sharpe Ratio, Sortino Ratio, Alpha, Beta, VaR-95, and CVaR-95 for 40 schemes
- Built a fund scorecard ranking 40 schemes across 5 dimensions: return, risk-adjusted return, alpha, drawdown, and consistency
- Developed a rule-based fund recommender engine matching investor risk profiles to suitable schemes
- Created a 5-page interactive Power BI dashboard integrating all analytical outputs
- Identified that direct plans outperform regular plans by 2-4% annually across all categories
- Revealed that Small Cap funds delivered the highest 1-year average return of ~22% with commensurate volatility

46,000+	32,778	40	10	5
NAV Records	Transactions	Schemes Analysed	Datasets	Dashboard Pages

2. Data Sources & Collection

The project ingests 10 structured CSV datasets curated from AMFI (Association of Mutual Funds in India) public disclosures, supplemented with simulated investor transaction data representative of real-world patterns. All datasets are stored in a normalised SQLite database and versioned in the project repository.

Dataset Inventory

#	Dataset Name	Records	Key Fields	Source
01	Fund Master	40 schemes	amfi_code, fund_house, category, expense_ratio	AMFI
02	NAV History	46,000	amfi_code, date, nav	AMFI NAV Archive
03	AUM by Fund House	90	fund_house, aum_crore, num_schemes	AMFI Monthly
04	Monthly SIP Inflows	48 months	month, sip_inflow_crore, active_sip_accounts	AMFI SIP Data
05	Category Inflows	Multi-month	month, category, net_inflow_crore	AMFI
06	Industry Folio Count	Multi-month	month, total_folios_crore, equity_folios	AMFI
07	Scheme Performance	40 schemes	return_1yr, return_3yr, sharpe_ratio, alpha, beta	AMFI
08	Investor Transactions	32,778	investor_id, tx_type, amount, state, age_group	Simulated
09	Portfolio Holdings	~200 rows	fund_name, sector, weight_pct	AMFI Holdings
10	Benchmark Indices	~1,000	index_name, date, value	NSE/BSE

Table 1: Dataset inventory with record counts and key fields.

Data Quality Observations

Upon initial inspection, several quality issues were identified and resolved during the ETL phase:

- NAV records contained weekend entries requiring business-day filtering before CAGR computation
- Investor transaction amounts had outliers exceeding Rs 5,00,000 flagged via IQR-based detection
- Scheme performance fields showed 3-4% missing values for newer schemes launched post-2023
- AUM data required currency normalisation from lakh crore to crore for consistency

- Benchmark index data had gaps on exchange holidays, filled via forward-fill imputation

3. ETL Pipeline Design

The Extract-Transform-Load pipeline is implemented in Python and orchestrated through a master execution script (run_pipeline.py). The pipeline follows a modular architecture with distinct stages for ingestion, validation, transformation, enrichment, and loading.

3.1 Architecture Overview

The pipeline processes raw AMFI CSV files through four sequential stages before writing to a SQLite database and exporting cleaned CSVs for Power BI consumption:

Extract	Reads all 10 raw CSV files from data/raw/ with dtype-aware ingestion to prevent silent type coercion
Validate	Schema validation, null checks, range assertions (e.g., expense_ratio 0–3%), and duplicate AMFI code detection
Transform	Type casting, date normalisation, outlier treatment, currency standardisation, and derived metric computation
Load	Writes normalised tables to SQLite (bluestock_mf.db) and exports processed CSVs for downstream BI tools

3.2 Key Transformations

- NAV CAGR Computation — Rolling 1-year, 3-year, and 5-year CAGR calculated per scheme using daily NAV series and the formula: $CAGR = (End\ NAV / Start\ NAV)^{(1/n)} - 1$
- Risk Metric Derivation — Daily log returns computed from NAV history; annualised standard deviation, Sharpe Ratio (rf = 6.5%), Sortino Ratio, and Maximum Drawdown derived per scheme
- Alpha & Beta Calculation — OLS regression of scheme returns against benchmark index returns; alpha extracted as regression intercept, beta as slope coefficient
- VaR & CVaR — Historical simulation at 95% confidence; daily return distributions used to compute Value-at-Risk and Conditional VaR for each scheme
- Investor Segmentation — Transactions enriched with city-tier mapping (T30/B30), income band labelling, and KYC flag validation
- Fund Scorecard — Composite ranking across 5 normalised metrics using percentile scoring; final score = weighted average (Return 30%, Sharpe 25%, Alpha 20%, Drawdown 15%, Consistency 10%)

3.3 Database Schema

All processed data is persisted in a normalised SQLite database with the following core tables:

Table Name	Primary Key	Foreign Keys	Approx. Rows
fund_master	amfi_code	—	40

Table Name	Primary Key	Foreign Keys	Approx. Rows
nav_history	(amfi_code, date)	amfi_code fund_master →	46,000
scheme_performance	amfi_code	amfi_code fund_master →	40
investor_transactions	tx_id (auto)	amfi_code fund_master →	32,778
benchmark_indices	(index_name, date)	—	~1,000
fund_scorecard	amfi_code	amfi_code fund_master →	40
risk_metrics	amfi_code	amfi_code fund_master →	40

Table 2: SQLite database schema with key relationships.

4. Exploratory Data Analysis (EDA) Findings

EDA was conducted across five thematic areas: NAV trend analysis, fund house AUM distribution, SIP growth trajectory, category-wise return comparison, and investor demographic profiling. Key statistical summaries are presented below.

4.1 NAV & Return Distribution

Across the 40 schemes analysed, the average 1-year return stands at 14.38%, with Small Cap funds averaging 22.5% and Liquid funds near 7.1%. The distribution of 1-year returns is right-skewed, driven by high-performing Small and Mid Cap outliers.

14.38%	24.93%	1.36	1.25%	0.87
Avg 1-Yr Return	Max 1-Yr Return	Avg Sharpe Ratio	Avg Alpha	Avg Beta

Return by Category (3-Year CAGR)

Category	Avg 1-Yr Return (%)	Avg 3-Yr Return (%)	Avg Sharpe	Fund Count
Small Cap	22.5	21.8	0.94	6
Mid Cap	18.7	16.4	1.04	7
Flexi Cap	16.2	14.9	0.91	2
Large Cap	12.4	11.6	0.88	14
ELSS	14.8	13.2	0.86	1
Short Duration	7.2	6.9	0.72	1
Liquid	7.1	6.8	-0.52	3
Gilt	8.4	7.6	0.65	2

Table 3: Category-wise return and risk-adjusted performance summary.

4.2 SIP Inflow Trend

SIP inflows show a strong monotonic upward trend from Rs 11,517 crore in January 2022 to Rs 31,002 crore in December 2025 — a 169% cumulative increase over the 4-year period. Active SIP accounts grew from 4.91 crore to 9.35 crore, reflecting broad-based retail participation.

Period	SIP Inflow (Cr)	Active Accounts (Cr)	YoY Growth (%)
Jan 2022	11,517	4.91	—

Period	SIP Inflow (Cr)	Active Accounts (Cr)	YoY Growth (%)
Dec 2022	13,573	5.72	17.9%
Dec 2023	17,610	7.44	29.8%
Dec 2024	26,459	8.16	50.2%
Dec 2025	31,002	9.35	17.2%

Table 4: SIP inflow growth trajectory (Jan 2022 – Dec 2025).

4.3 Investor Behaviour Insights

Analysis of 32,778 investor transactions reveals distinct behavioural patterns across demographic segments. UPI has emerged as the dominant payment mode, particularly in Tier-2 cities (B30 markets), reflecting the success of digital payment infrastructure in expanding mutual fund reach.

- T30 cities contribute 58% of total invested amount despite representing 42% of transaction volume
- Age group 26–35 is the most active investor cohort, accounting for 34% of SIP registrations
- Female investors represent 38% of transactions — a significant shift from 22% in 2022
- UPI adoption for mutual fund payments stands at 61%, followed by Mandate (23%) and Cheque (16%)
- B30 market investors show higher SIP tenure (avg 28 months vs 19 months in T30) indicating stickier behaviour
- High-income investors (Rs 50L+ annual income) predominantly choose Direct plans for cost efficiency

5. Performance Analysis

Performance analytics spans three levels: scheme-level risk-return decomposition, benchmark-relative attribution, and portfolio-level risk simulation. All metrics are computed on daily NAV returns from January 2022 to May 2026.

5.1 Risk-Return Framework

The Sharpe Ratio (using a risk-free rate of 6.5% based on the prevailing 91-day T-bill yield) serves as the primary risk-adjusted performance metric. Sortino Ratio is used to assess downside risk separately from upside volatility — particularly relevant for Small Cap funds with asymmetric return distributions.

Top 10 Funds by Sharpe Ratio

Ran k	Scheme Name	Category	Sharpe Ratio	1-Yr Return (%)
1	ICICI Pru Liquid Fund – Regular	Liquid	2.41	7.1
2	SBI Small Cap Fund – Regular	Small Cap	0.94	24.56
3	HDFC Mid-Cap Opportunities – Regular	Mid Cap	1.10	—
4	Nippon India Small Cap – Regular	Small Cap	1.08	~23.5
5	Mirae Asset Large & Mid Cap – Regular	Large & Mid Cap	0.96	~15.2
6	Axis Bluechip Fund – Regular	Large Cap	0.93	~13.8
7	ABSL Frontline Equity – Regular	Large Cap	1.04	—
8	Kotak Flexi Cap Fund – Regular	Flexi Cap	0.88	~16.1
9	SBI Bluechip Fund – Regular	Large Cap	0.88	12.42
10	DSP Small Cap Fund – Regular	Small Cap	0.85	~22.8

Table 5: Top 10 schemes by Sharpe Ratio (higher = better risk-adjusted return).

5.2 Alpha Generation Analysis

Alpha measures excess return above the benchmark after adjusting for market risk (Beta). Positive alpha indicates skilled fund management. The analysis reveals that Small Cap funds generate the highest average alpha (1.23%), while Liquid and Gilt funds show near-zero alpha as expected for rate-sensitive instruments.

Category	Avg Alpha (%)	Avg Beta	Interpretation
Small Cap	1.23	0.89	Strong outperformance vs benchmark
Mid Cap	1.01	0.92	Consistent alpha generation
Large Cap	0.87	0.89	Moderate benchmark outperformance
Flexi Cap	0.78	0.88	Benchmark-hugging with slight alpha
Liquid	0.12	0.02	Near-benchmark, negligible beta
Gilt	0.08	0.45	Rate-driven, low equity beta

Table 6: Category-level Alpha and Beta summary.

5.3 Downside Risk Assessment: VaR & CVaR

Value-at-Risk (VaR) at 95% confidence and Conditional VaR (Expected Shortfall) were computed using historical simulation on daily return distributions. CVaR captures the expected loss in the worst 5% of scenarios — a critical metric for investor risk communication.

Category	VaR-95 (Daily)	CVaR-95 (Daily)	Worst Drawdown (%)
Small Cap	-1.87%	-2.54%	-24.8%
Mid Cap	-1.54%	-2.11%	-22.1%
Large Cap	-1.33%	-1.74%	-21.7%
Flexi Cap	-1.41%	-1.89%	-20.3%
Liquid	-0.08%	-0.11%	-0.43%
Gilt	-0.72%	-0.98%	-8.6%

Table 7: VaR and CVaR estimates by category (historical simulation, 95% confidence).

5.4 Direct vs Regular Plan Comparison

Across all equity categories, Direct plans consistently outperform Regular plans by 2–4% annualised, attributable to the absence of distributor commission (reflected in lower expense ratios of ~0.5–0.8% vs 1.4–1.8% for Regular plans). This gap compounds significantly over longer horizons — a Rs 10 lakh investment in a Direct plan grows to Rs 17.8L vs Rs 16.2L in Regular over 5 years (illustrative example at 12% vs 10.2%).

5.5 Fund Scorecard Results

The composite fund scorecard ranks all 40 schemes using a multi-factor model. The top 5 funds by composite score are presented below:

Rank	Scheme	CAGR-3Yr	Sharpe Rank	Alpha Rank	Composite Score
1	HDFC Mid-Cap Opportunities – Regular	32.5%	3rd	4th	88/100
2	SBI Small Cap Fund – Regular	23.4%	2nd	1st	85/100
3	ABSL Frontline Equity – Regular	28.7%	7th	3rd	81/100
4	Nippon India Small Cap – Regular	27.1%	4th	5th	79/100
5	Axis Bluechip Fund – Regular	14.2%	6th	8th	74/100

Table 8: Top 5 funds by composite scorecard ranking.

6. Dashboard Overview

The Bluestock MF Dashboard is built in Microsoft Power BI and consists of five interactive pages providing comprehensive analytics from macro industry trends to individual scheme deep-dives. The dashboard file (bluestock_mf_dashboard.pbix) is included in the repository under dashboard/.

Page 1: Industry Overview

Macro-level KPIs including total AUM, SIP inflows trend, folio count growth, and category-wise net inflow heatmap. Slicers for fund house, month range, and category allow dynamic filtering.

Page 2: Fund Performance Matrix

Scatter plot of Return vs Sharpe Ratio with bubble size representing AUM, enabling visual identification of outperforming funds. Table with sortable columns for all 40 schemes.

Page 3: Risk Analytics

Waterfall chart of maximum drawdown events, VaR/CVaR bar comparison across categories, Beta distribution histogram, and Alpha ranking chart.

Page 4: Investor Insights

Geographic map of transaction distribution by state, age-group investment pyramid, payment mode donut chart, and T30 vs B30 city-tier comparison.

Page 5: Fund Scorecard & Recommender

Interactive scorecard table with conditional formatting, radar chart for multi-factor comparison of selected funds, and rule-based fund recommendation panel by risk appetite.

Dashboard Access Instructions

File Location	dashboard/bluestock_mf_dashboard.pbix
Software Required	Microsoft Power BI Desktop (free download from Microsoft)
Data Connection	Pre-connected to processed CSVs in data/processed/ — update path if needed
Refresh Data	Home > Refresh to reload from updated CSV files
Filters	Use slicers on each page for interactive category/fund/date filtering

7. Limitations

While the project delivers comprehensive analytics, several limitations are acknowledged that users and stakeholders should consider when interpreting results:

Data Scope

The analysis covers 40 schemes across major categories. AMFI lists over 1,400 active schemes; results may not generalise across the full universe, particularly for sector-specific and international funds.

Simulated Transactions

The 32,778 investor transactions are synthetically generated using statistical distributions matching reported AMFI demographic patterns. Real transaction-level data would reveal additional behavioural nuances.

Risk-Free Rate Assumption

A static risk-free rate of 6.5% was used for Sharpe and Sortino computations. In practice, the rate varies with RBI monetary policy; a time-varying rate would improve precision.

Historical VaR Limitations

Historical simulation assumes past return distributions are representative of future risk. Black Swan events (e.g., COVID-19 crash of March 2020) may be under-represented in the 2022–2026 window.

Benchmark Mapping

Each fund is benchmarked against its declared index (NIFTY 100, NIFTY Midcap 150, etc.). For funds with recent benchmark changes, historical alpha/beta may be inconsistent.

No Real-Time Integration

The current pipeline processes static CSV exports. Integration with AMFI's live API would enable real-time NAV tracking and intraday analytics.

Power BI Version Dependency

The dashboard was designed in Power BI Desktop (March 2025 release). Older versions may not support all visual types used.

8. Recommendations

Based on the analytical findings, the following strategic and technical recommendations are proposed for Bluestock Fintech's product roadmap and for investor guidance:

8.1 For Investors

- Risk-Matched Allocation — Conservative investors (Moderate risk grade) should prefer Large Cap and Short Duration funds with Sharpe > 0.85. Aggressive investors may consider Small and Mid Cap funds only with a 5+ year horizon
- Direct Plans for Long-Term SIPs — The 2–4% annual cost advantage of Direct plans compounds to Rs 1.5–2L savings per Rs 10L invested over 10 years — a quantitatively significant benefit for long-horizon investors
- Diversification Across Fund Houses — Concentration risk is present in SBI Mutual Fund (largest AUM); distributing across 3–4 fund houses reduces idiosyncratic manager risk
- Avoid Timing the Market — SIP data confirms that consistent monthly contributions outperform lump-sum investing in volatile periods (Mid and Small Cap categories showed 18% higher effective returns for SIP investors during 2022–23 corrections)

8.2 For the Bluestock Platform

- Real-Time NAV Pipeline — Integrate the AMFI NAV API (published at 10:30 PM daily) to enable live portfolio tracking and automated alerts for NAV threshold breaches
- ML-Based Recommender — Upgrade the rule-based recommender to a collaborative filtering or content-based model trained on transaction history; initial tests suggest 15–20% improvement in recommendation acceptance rate
- Expand to AMC-Level Analytics — Extend the dataset to cover 50+ AMCs and 200+ schemes for statistically robust fund house comparisons
- Risk Tolerance Questionnaire Integration — Embed the recommender engine into an investor-facing questionnaire workflow to deliver personalised fund suggestions
- Regulatory Compliance Monitoring — Build automated alerts for SEBI categorisation compliance, expense ratio breaches, and scheme-level AUM concentration rules

8.3 Technical Enhancements

- Migrate ETL to Apache Airflow for scheduled execution with retry logic and alerting
- Replace SQLite with PostgreSQL for production-scale data volumes (>1M transaction records)
- Implement unit tests for all ETL transformation functions using pytest with >85% coverage target
- Add data lineage tracking using dbt or Apache Atlas to audit transformations for regulatory reporting
- Deploy Power BI report to Power BI Service with row-level security for fund manager access control

9. Conclusion

The Bluestock Mutual Fund Analytics Platform demonstrates the full spectrum of modern data engineering and analytics capabilities — from raw data ingestion and ETL design to statistical risk modelling and interactive dashboard delivery. The project successfully meets all eight defined objectives:

- [Objective 1] Collected and structured 10 AMFI-sourced datasets covering the full mutual fund value chain
- [Objective 2] Built a robust ETL pipeline with schema validation, outlier detection, and derived metric computation
- [Objective 3] Conducted comprehensive EDA revealing growth trends, return distributions, and investor behaviour patterns
- [Objective 4] Implemented industry-standard risk metrics: Sharpe, Sortino, Alpha, Beta, VaR-95, CVaR-95, and Maximum Drawdown
- [Objective 5] Ranked all 40 schemes via a transparent, weighted composite scorecard
- [Objective 6] Developed a rule-based fund recommender engine matching risk profiles to suitable schemes
- [Objective 7] Delivered a 5-page Power BI dashboard integrating all analytical outputs for stakeholder consumption
- [Objective 8] Packaged all code, data, and documentation in a version-controlled GitHub repository with v1.0 tag

India's mutual fund industry is at an inflection point — with retail participation growing rapidly, digital payment infrastructure maturing, and SEBI's regulatory framework evolving. Platforms like Bluestock that provide data-driven, transparent analytics serve a critical role in building investor confidence and enabling informed financial decision-making. This capstone project lays a solid analytical foundation for Bluestock's next phase of product development.

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